CCCA Meta-Atlas · Embryonic-Program Cell-State Analysis

Geistlinger et al. 2020 · High-Grade Serous Ovarian Carcinoma · 10x · 5,914 malignant cells (5 patients)

GO / NO-GO GATES

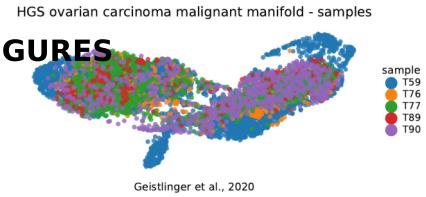
GATE 1 GO — Mixing + continuity: 5 patients share ONE continuous manifold (single kNN component; iLISI 1.87/5, limited by 54% T59 dominance). θ=5 resolves a T59-only under-integration lobe seen at θ=2. Judged on the embedding.

GATE 2 GO — Embryonic pattern present, non-branching: pSom↔Crest single inverse axis (r=−0.95 smoothed); Gast co-varies with pSom (+0.61), both oppose Crest. Crest is the terminal gradient (crest-high decile at pseudotime 0.98). Not a proliferation artifact — pSom↔Crest holds in non-cycling cells (r=−0.62). Crest at the terminus, not the Gast arm → no flag.

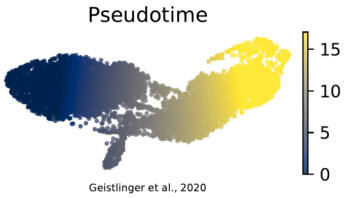
GATE 3 NA — Branching-only. No bifurcation (fold-free 2-tip curve; PPT branch-node counts unstable 3–8, no reproducible fork; Gast⊥pSom, r≈0) → no hub cross-repression test. Non-branching case, per §1–2.

STEP FIGURES

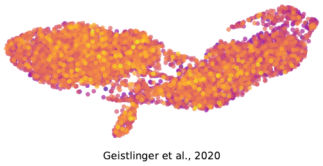
01 · Samples (Gate 1 integration)



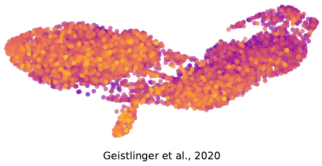
02 · Pseudotime manifold



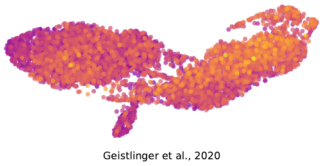
03 · Gastrulation



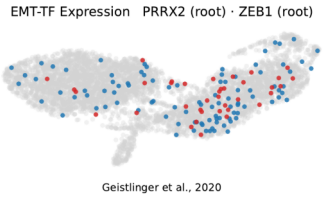
03 · Somitogenesis (pSom)



03 · Neural crest



04 · EMT-TF expression



PARAMETER CHOICES & RATIONALE

Min-cell threshold	≥100 (floor) → 5/5 samples kept	Multiplicative-gap cut lands at the HEAD (3180→1298), not a sparse tail → premise unmet → precedent floor.
n_pcs / integration	20 PCs · Harmony θ=5	Scree elbow ~5 rejected as under-powering (Chen n_pcs=6 failure); θ=5 merges the T59 under-integration lobe.
Island removal	radius-CC, 17-cell speck	θ=5 removes only a true speck; the 740-cell θ=2 lobe (95% T59) was under-integration, not a lineage — kept.
Geometry / trajectory	Non-branching · scFates curve	Fold-free 2-tip curve (self-approach 2.97). pSom/Crest inverse gradient; no Gast↔pSom divergence.
Root	Gast/pSom-high tip (Crest-low)	Neural crest is developmentally late → root at the Gast/pSom origin, terminus at Crest.
Subtype annotation	TCGA + 3CA MPs + Leiden	HGSOC malignant cells ~85% DIF across all lobes → continuous gradient, not discrete molecular subtypes.
EMT-TFs shown	PRRX2 · ZEB1	Detection-gated wide panel (FOXC/PRRX incl.); all detectable factors root/mesenchymal-enriched, opposite Crest.

VERDICT The embryonic-program pattern REPRODUCES as the non-branching case: one continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high origin, with Crest an inverse terminal gradient — a strict linear subset of the Puram branching hub. G1 GO · G2 GO (non-branching) · G3 NA. No hub / no cross-repression r (branching-only).